



**Technological Institute of the Philippines
College of Computer Studies**

Final Project Proposal

Submitted By:
SALVADOR, LOUIS ABRAHAM

Submitted To:
FRANCIS R. CARABUENA

Proposed Title:

1.)Gaming Habits Affecting Academic Performance

Problem Statement

This proposed title aims to analyze the relationship between students' gaming habits and their academic performance.

Relevance:

Gaming is a major recreational activity among students. Understanding how it impacts performance and spending helps educators, parents, and policymakers support balanced digital engagement.

Proposed Dataset:

<https://zenodo.org/records/15633261>

<https://huggingface.co/datasets/ismail31415/uniGame>

<https://www.nature.com/articles/s41597-025-05614-4?>

Initial Analytical Approach:

EDA: Explore patterns between gaming hours and GPA,

Predictive: Use regression or classification to predict academic performance based on gaming behavior

Expected Insights: Identify whether excessive gaming correlates with lower academic performance, detecting the threshold levels of gaming affecting outcomes

Justification:

This topic explores how students' gaming behavior impacts their academic performance. Understanding this relationship can help educators and parents make informed decisions about screen time. The dataset includes gaming frequency, time spent, and academic scores, allowing predictive analysis. Insights may reveal patterns that guide interventions to improve student outcomes.

Proposed Title:

2.)Health Disease Prediction

Problem Statement

This proposed title aims to predict the likelihood of patients developing specific diseases based on health indicators and lifestyle factors

Relevance:

Early detection of diseases can save lives and reduce healthcare costs. Predictive analytics can assist doctors in prioritizing care and preventive measures.

Proposed Dataset:

<https://www.kaggle.com/datasets?topic=healthDataset>

https://baselight.app/u/kaggle/dataset/prosperchuks_health_dataset?

https://baselight.app/u/kaggle/dataset/alexteboul_diabetes_health_indicators_dataset

Initial Analytical Approach:

EDA: Examine distributions of health metrics, correlations with disease incidence

Predictive: Classification models (Decision Tree, Logistic Regression) to predict disease probability

Expected Insights: Identify high-risk individuals and key health factors influencing disease development

Justification:

Predicting the risk of diseases like diabetes, heart disease, or kidney disease is crucial for early intervention and preventive care. Health datasets with clinical and lifestyle indicators enable classification modeling. This project can help identify high-risk individuals and provide actionable insights for healthcare planning. The predictive model may improve awareness and support timely treatment.

Proposed Title:

3.)Oil Price Analysis

Problem Statement

This proposed title aims to forecast global crude oil price volatility and also aims to analyze the impact of geopolitical tensions involving Iran and disruptions in the Strait of Hormuz on global oil prices.

Relevance:

The analysis of oil prices is relevant as it provides crucial insights for risk management, strategic economic planning, and time series forecasting amid global price volatility and geopolitical tensions.

Proposed Dataset:

<https://www.kaggle.com/datasets/alitaqishah/iran-war-oil-shock-2026-brent-and-gas-tracker>

<https://www.kaggle.com/datasets/kavyadhyani/global-oil-prices-and-geopolitical-events>

<https://www.kaggle.com/datasets/meharshanali/middle-east-economy-and-oil-prices-1990-2024>

Initial Analytical Approach:

EDA: Analyze price trends before, during, and after the crisis; visualize volatility spikes

Predictive: Time series forecasting to predict future oil price trends

Expected Insights: Quantify the effect of conflicts on oil prices, identify patterns in supply disruptions

Justification:

The recent crisis in the Strait of Hormuz has caused volatility in global oil prices.

Analyzing historical and recent price data helps understand the impact of geopolitical events on energy markets. Insights can guide policymakers, investors, and businesses in risk assessment. Time series and predictive modeling can reveal patterns and forecast potential market shifts.

Proposed Title:

4.)Car Price Prediction

Problem Statement

This proposed title aims to predict car prices based on vehicle features, while considering macroeconomic factors such as oil price fluctuations.

Relevance:

Car pricing affects consumers and dealers; understanding the impact of fuel price volatility helps with market planning and economic forecasting.

Proposed Dataset:

<https://www.kaggle.com/datasets/sidharth178/car-prices-dataset?>

<https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset?>

Initial Analytical Approach:

EDA: Examine relationships between car features and prices

Predictive: Regression models to estimate car prices based on features and external factors (oil prices)

Expected Insights: Determine key factors affecting car prices and quantify the effect of energy market fluctuations

Justification:

Car prices are influenced by multiple factors, including demand, features, and oil prices. This topic allows analysis of market trends and prediction of vehicle prices using structured datasets. Understanding pricing patterns helps buyers, sellers, and businesses make informed decisions. Linking it to oil prices adds relevance to the current economic context.

Proposed Title:

5.)Flight Delay Prediction

Problem Statement

This proposed title aims to predict flight delays using historical flight data, airline operations, and external factors such as oil price volatility.

Relevance:

Flight delays affect passengers, airlines, and the economy. Understanding contributing factors can improve scheduling and operational efficiency.

Proposed Dataset:

<https://www.kaggle.com/datasets/hrishitpatil/flight-data-2024?>

<https://www.kaggle.com/datasets/underscore/flight-delay-and-causes>

<https://www.kaggle.com/datasets/sriharshaeedala/airline-delay?>

Initial Analytical Approach:

EDA: Explore delay patterns by airline, route, and external factors

Predictive: Classification/regression models to predict flight delay probability or duration

Expected Insights: Identify factors driving delays and the influence of fuel prices on operational performance

Justification:

Airline delays affect passengers, logistics, and airline operations. Predicting delays using historical flight data and factors like weather and oil prices can improve scheduling and planning. This project helps identify patterns that cause delays and optimize flight management. Integrating oil price data highlights how global events influence operational efficiency.

Reflection:

Choosing a topic was challenging because I wanted to find problems that are both interesting and have accessible, reliable datasets. I realized that data-driven problem solving requires balancing relevance, data availability, and the potential for actionable insights. This process taught me that even simple questions can become complex when considering multiple influencing factors. Overall, I gained a deeper appreciation for how analytics can uncover patterns and inform decisions in diverse areas like education, health, and economics.